2 Classical statistical physics – solutions

The problems are roughly in order of difficulty. Those indicated with ♡ are ones which are highly recommended. Any labelled with ♣ are the hardest ones, which might only occur as a "sting in the tail" at the end of a long examination question.

♡*Problem 2.1* **Average potential energy for a harmonic oscillator**

(i) The partition function is (using the separability of q and p)

$$Z(T) = \frac{1}{\mathsf{h}} \int_{-\infty}^{\infty} \mathrm{d}q \ \exp\left(-\frac{1}{2}\beta K q^2\right) \int_{-\infty}^{\infty} \mathrm{d}p \ \exp\left(-\beta \frac{p^2}{2m}\right) = \frac{1}{\mathsf{h}} Z_q(T)\, Z_p(T) \ .$$

(ii) Use $\int_{-\infty}^{\infty} \mathrm{d}x \ \mathrm{e}^{-\alpha x^2} = \sqrt{\pi/\alpha}$,

$$Z(T) = \frac{1}{\mathsf{h}} \sqrt{\frac{2\pi k_{\mathrm{B}} T}{K}} \sqrt{2\pi m k_{\mathrm{B}} T} = \frac{k_{\mathrm{B}} T}{\hbar \omega_0} \ ,$$

where $\omega_0 = \sqrt{K/m}$ is the natural frequency of the oscillator.

(iii) The probability distribution $p(q,p)$ is

$$p(q,p) = \frac{1}{\mathsf{h}} \frac{\exp\left(-\frac{1}{2}\beta\left[K q^2 + \frac{p^2}{m}\right]\right)}{Z(T)} \ .$$

(iv) The average potential energy is

$$\begin{aligned}\langle V(q)\rangle &= \frac{\frac{1}{\mathsf{h}} \int_{-\infty}^{\infty} \mathrm{d}q \ \frac{1}{2} K q^2 \exp\left(-\frac{1}{2}\beta K q^2\right) \int_{-\infty}^{\infty} \mathrm{d}p \ \exp\left(-\beta \frac{p^2}{2m}\right)}{Z(T)} \\ &= \frac{\frac{1}{\mathsf{h}} \int_{-\infty}^{\infty} \mathrm{d}q \ \frac{1}{2} K q^2 \exp\left(-\frac{1}{2}\beta K q^2\right) Z_p(T)}{\frac{1}{\mathsf{h}} Z_q(T)\, Z_p(T)} \ .\end{aligned}$$

Let

$$y^2 = \tfrac{1}{2}\beta K q^2 \quad \Rightarrow \quad \mathrm{d}q = \sqrt{\frac{2 k_{\mathrm{B}} T}{K}} \mathrm{d}y \ .$$

Then using expression for $Z_q(T)$ and $\int_{-\infty}^{\infty} \mathrm{d}y \ y^2 \mathrm{e}^{-y^2} = \sqrt{\pi}/2$ (e.g. by Feynman trick – see Maths 1B), we see

$$\begin{aligned}\langle V(q)\rangle &= \beta^{-3/2} \sqrt{2/K} \frac{\int_{-\infty}^{\infty} \mathrm{d}y \ y^2 \mathrm{e}^{-y^2}}{Z_q(T)} \\ &= \frac{\beta^{-3/2} \sqrt{2/K} \sqrt{\pi}}{2\sqrt{2\pi k_{\mathrm{B}} T/K}} \\ &= \tfrac{1}{2} k_{\mathrm{B}} T \ ,\end{aligned}$$

as expected from equipartition.

♡*Problem 2.2* **Energy fluctuations**
Consider the expression for the Gaussian integral:

$$\int_{-\infty}^{\infty} \mathrm{d}x\ \mathrm{e}^{-\alpha x^2} = \sqrt{\frac{\pi}{\alpha}}\ ,$$

so, using the Feynman trick, differentiating both sides with respect to α,

$$\begin{aligned} -\frac{\mathrm{d}}{\mathrm{d}\alpha}\int_{-\infty}^{\infty} \mathrm{d}x\ \mathrm{e}^{-\alpha x^2} = -\int_{-\infty}^{\infty} \mathrm{d}x\ \frac{\partial}{\partial\alpha}\mathrm{e}^{-\alpha x^2} &= \int_{-\infty}^{\infty} \mathrm{d}x\ x^2\mathrm{e}^{-\alpha x^2} \\ \text{and}\quad -\frac{\mathrm{d}}{\mathrm{d}\alpha}\sqrt{\frac{\pi}{\alpha}} &= \frac{1}{2}\sqrt{\frac{\pi}{\alpha^3}} \\ \Rightarrow\quad \int_{-\infty}^{\infty} \mathrm{d}x\ x^2\mathrm{e}^{-\alpha x^2} &= \frac{1}{2}\sqrt{\frac{\pi}{\alpha^3}}\ . \end{aligned} \tag{2.1}$$

Consider the expectation value, using $Z(T,V) = \int \mathrm{D}E\ W(E)\mathrm{e}^{-\beta E}$,

$$\langle (E-U)^2\rangle = \frac{\int \mathrm{d}E\ W(E)(E-U)^2\mathrm{e}^{-\beta E}}{Z}\ .$$

Now use the (very, very good) approximation for large N, Eqn. (4.4) from section 4:

$$Z(T,V) = \int \mathrm{d}E P_{N\to\infty}(E,V) = \mathrm{e}^{-\beta F(T,V,N)}\int \mathrm{d}E\ \exp\left(-\frac{(E-U)^2}{2k_{\mathrm{B}}T^2C_V}\right)\ ,$$

to define average:

$$\begin{aligned} \langle (E-U)^2\rangle_{N\to\infty} &= \frac{\int \mathrm{d}E(E-U)^2 P_{N\to\infty}(T,V)}{\int \mathrm{d}E P_{N\to\infty}(T,V)} \\ &= \frac{\int \mathrm{d}E\ (E-U)^2 \exp\left(-\frac{(E-U)^2}{2k_{\mathrm{B}}T^2C_V}\right)}{\int \mathrm{d}E\ \exp\left(-\frac{(E-U)^2}{2k_{\mathrm{B}}T^2C_V}\right)}\ . \end{aligned}$$

Now use the Gaussian integral result and that derived at start of question with $\alpha = 1/(2k_{\mathrm{B}}T^2C_V)$ to obtain

$$\langle (E-U)^2\rangle_{N\to\infty} = \frac{1}{2}\left.\frac{\sqrt{\pi/\alpha^3}}{\sqrt{\pi/\alpha}}\right|_{\alpha^{-1}=2k_{\mathrm{B}}T^2C_V} = k_{\mathrm{B}}T^2C_V\ .$$

Noting that the specific heat is an extensive variable, the fractional fluctuation in energy (i.e. root mean square of fluctuation compared to average energy)

$$\frac{\sqrt{\langle (E-U)^2\rangle}}{U} \propto \frac{\sqrt{N}}{N} = \frac{1}{\sqrt{N}}\ .$$

This shows fluctuations are negligible if

$$N \sim N_{\mathrm{A}} = 10^{23} \Rightarrow \frac{1}{\sqrt{N}} \sim 10^{-12} \ll 1\ .$$

♡*Problem 2.3* **Pressure and internal energy for ideal gas**
In lectures we derived the free energy of the ideal gas:

$$F(T,V,N) = -Nk_{\text{B}}T\left\{\ln\left(\frac{v}{\lambda_{\text{T}}^3}\right)+1\right\} \ .$$

where

$$v = \frac{V}{N} \quad \text{and} \quad \lambda_{\text{T}} = \left(\frac{\hbar^2}{2\pi m k_{\text{B}}T}\right)^{1/2} \ .$$

(i) If we take the general canonical partition function:

$$Z(T,V,N) = \int \mathrm{d}E\ W(E)\mathrm{e}^{-\beta E} \ ,$$

with, as usual, $F = -k_{\text{B}}T\ln Z$, then show

$$U = \langle E\rangle = \frac{\partial(\beta F)}{\partial\beta} \ .$$

This complements the derivation thermodynamically at the beginning of section 4.

$$\begin{aligned}
\frac{\partial(\beta F)}{\beta} &= \frac{\partial}{\partial\beta}\left[-\underbrace{k_{\text{B}}T\beta}_{\text{cancel}}\ln Z\right] \\
&= -\frac{1}{Z}\frac{\partial Z}{\partial\beta} \\
&= -\frac{1}{Z}\int \mathrm{d}E\ W(E)(-E)\mathrm{e}^{-\beta E} \\
&= +\int \mathrm{d}E\ W(E)\,E\,\underbrace{\frac{\mathrm{e}^{-\beta E}}{Z}}_{P(E)} \\
&= \int \mathrm{d}E\ W(E)\,E\,P(E) \\
&= \langle E\rangle \\
&= U \ .
\end{aligned}$$

(ii) Use the expression in last item to derive the internal energy, U of the ideal gas and hence the specific heat at constant volume, C_V.

$$\begin{aligned}
U = \frac{\partial(\beta F)}{\partial\beta} &= -N\frac{\partial}{\partial\beta}\left\{\ln\left(\frac{v}{\lambda_{\text{T}}^3}\right)+1\right\} \\
&= N\frac{\partial}{\partial\beta}\ln\left(\lambda_{\text{T}}^3\right) \\
&= \frac{3}{2}N\frac{\partial}{\partial\beta}\ln\left(\frac{\hbar^2}{2\pi m}\beta\right) \\
&= \frac{3}{2}N\frac{1}{\beta} \\
&= \frac{3}{2}Nk_{\text{B}}T \ .
\end{aligned}$$

As expected!
To obtain C_V is straightforward

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = \frac{3}{2}Nk_B = \frac{3}{2}R \ .$$

Again as expected.

♡*Problem 2.4* **Spin-1/2 paramagnet**

In the lectures we derive the (magnetic Gibbs) free energy, $G_M(T, H, N)$, of the spin-1/2 paramagnet, with magnetic moment (including a factor of μ_0), μ in an external field H:

$$G_M(T, H, N) = -k_B T \ln Z = -Nk_B T \ln\left[2\cosh(\beta\mu H)\right] \ .$$

Derive the entropy of the spins, S.

$$\begin{aligned}
S = -\left(\frac{\partial G}{\partial T}\right)_V &= N\frac{\partial}{\partial T}\left\{k_B T \ln\left[2\cosh(\beta\mu H)\right]\right\} \\
&= Nk_B\left\{\ln\left[2\cosh(\beta\mu H)\right] + T \underbrace{\frac{\mathrm{d}\beta}{\mathrm{d}T}\frac{\partial}{\partial\beta}}_{\text{change variables for convenience}} \ln\left[2\cosh(\beta\mu H)\right]\right\} \\
&= Nk_B\left\{\ln\left[2\cosh(\beta\mu H)\right] - \frac{1}{k_B T}\frac{2\sinh(\beta\mu H)}{2\cosh(\beta\mu H)}\mu H\right\} \\
\text{(As } T\to\infty\text{)} \simeq Nk_B &\left\{\ln 2 - \frac{1}{2}\left(\frac{\mu H}{k_B T}\right)^2\right\} \\
&\simeq Nk_B \ln 2 \ .
\end{aligned}$$

Which is reasonable: $\ln 2$ entropy per spin, as there are two states, $\uparrow, \downarrow$, and temperature has overwhelmed the influence of the field energy. I.e. "high T" is $k_B T \gg \mu H$.

Problem 2.5 **Spin-1 paramagnet**
Let there be N spin-1 atoms, with their spin state denoted $\sigma_n = -1, 0, 1$ for $1 \le n \le N$. It has an associated magnetic moment, μ (which incorporates a factor of μ_0), and has interaction with an external magnetic field, H:

$$\mathcal{H} = -\mu H \sum_{n=1}^{N} \sigma_n \ .$$

(i) Write down the variable expression the magnetisation for this problem:

$$\frac{m}{\mu} = \sum_{n=1}^{N} \sigma_n \ .$$

This is identical to the spin 1/2 case, but the σ's now have three possible values: $\sigma_n = 0, \pm 1$.

(ii) Write down the canonical partition function, $Z(T,H,N)$ for the N spins.
Firstly we write down the number of states, $W(E)$,

$$W(E) = \sum_{\sigma_1=-1}^{1}\sum_{\sigma_2=-1}^{1}\cdots\sum_{\sigma_N=-1}^{1}\delta\left(E-\left[-\mu H\sum_{n=1}^{N}\sigma_n\right]\right)$$

(Defining notation for overall sum)
$$= \sum_{\{\sigma_n\}}\delta\left(E-\left[-\mu H\sum_{n=1}^{N}\sigma_n\right]\right) .$$

Then we substitute this expression into:

$$\begin{aligned}
Z(T,H,N) &= \int \mathrm{d}E\; W(E)\,\mathrm{e}^{-\beta E} \\
&= \int \mathrm{d}E\; \sum_{\{\sigma_n\}}\delta\left(E-\left[-\mu H\sum_{n=1}^{N}\sigma_n\right]\right)\mathrm{e}^{-\beta E} \\
&= \sum_{\{\sigma_n\}}\exp\left\{+\beta\mu H\sum_{n=1}^{N}\sigma_n\right\} && \text{(Let } \delta\text{-function eliminate integral.)} \\
&= \sum_{\{\sigma_n\}}\prod_{n=1}^{N}\exp\{\beta\mu H\sigma_n\} && \text{(Use the sum to product trick.)} \\
&= \sum_{\sigma_1=-1}^{1}\sum_{\sigma_2=-1}^{1}\cdots\sum_{\sigma_N=-1}^{1}\prod_{n=1}^{N}\exp\{\beta\mu H\sigma_n\} \\
&= \sum_{\sigma_1=-1}^{1}\mathrm{e}^{\beta\mu\sigma_1 H}\sum_{\sigma_2=-1}^{1}\mathrm{e}^{\beta\mu\sigma_2 H}\cdots\sum_{\sigma_N=-1}^{1}\mathrm{e}^{\beta\mu\sigma_N H} \\
&= [\underbrace{\mathrm{e}^{\beta\mu H}}_{\sigma=1}+\underbrace{1}_{\sigma=0}+\underbrace{\mathrm{e}^{-\beta\mu H}}_{\sigma=-1}]^N && \text{(Each sum has identical value.)} \\
&= [1+2\cosh(\beta\mu H)]^N
\end{aligned}$$

Notice the spins are non-interacting so we find (as before for non-interacting systems) a product of the partition functions for each spin expressing the total partition function. Thus the magnetic Gibbs free energy is

$$\begin{aligned}
G_{\mathrm{M}}(T,H,N) &= -k_{\mathrm{B}}T\ln Z(T,H,N) \\
&= -k_{\mathrm{B}}TN\ln[1+2\cosh(\beta\mu H)] .
\end{aligned}$$

(iii) Calculate the average magnetisation, m and isothermal susceptibility, χ_T for the set of spins-1.

$$\begin{aligned}
m &= -\frac{\partial G_{\mathrm{M}}}{\partial H} \\
&= +k_{\mathrm{B}}TN\frac{\partial}{\partial H}\ln[1+2\cosh(\beta\mu H)] \\
&= k_{\mathrm{B}}TN\frac{2\sinh(\beta\mu H)}{1+2\cosh(\beta\mu H)}\beta\mu \\
&= N\mu\frac{2\sinh(\beta\mu H)}{1+2\cosh(\beta\mu H)} .
\end{aligned}$$

The limits at large and small T (or small and large β),

$$\begin{aligned} m(T) &\underset{T\to 0}{\sim} N\mu \frac{\mathrm{e}^{\beta\mu H}}{\mathrm{e}^{\beta\mu H}} \\ &= N\mu \\ m(T) &\underset{T\to\infty}{\sim} N\mu \frac{2}{3}\beta\mu H \\ &= \frac{2}{3}(N\mu)\,\frac{\mu H}{k_{\mathrm{B}}T} \ . \end{aligned}$$

So as $T \to 0$, the spins are fully magnetised (all spins are $\uparrow$), and as $T \to \infty$, the magnetisation tends to zero, proportional to $1/T$.

Problem 2.6 **Elastic band**

Consider N rods (which we assume are massless and can move through each other) of unit length joined together, so that if the assembly is stretched out it has a length N. See Fig. (2.1).

One end of the set is attached to a support and the other hand has a mass M attached.

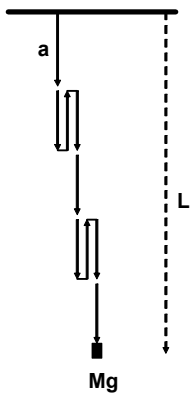

Figure 2.1: Chain of rods hanging under the influence of a weight.

Note that the distance between the two ends of the chain is L, *measured parallel to the gravitational force* i.e. "down", and this is related to the number of up, $n_\uparrow$, and down, $n_\downarrow$, rods:

$$L = (n_\downarrow - n_\uparrow) \ .$$

(i) Write down the number of configurations with L, $W(N, L)$, $W(n_\uparrow, n_\downarrow)$ or $W(N, n_\uparrow)$

$$\begin{aligned} W(N,L) &= \frac{N!}{\left[\frac{1}{2}(N+L)\right]!\ \left[\frac{1}{2}(N-L)\right]!} \\ \Rightarrow \quad W(n_\uparrow, n_\downarrow) &= \frac{(n_\uparrow + n_\downarrow)!}{(n_\uparrow)!, (n_\downarrow)!} \\ \Rightarrow \quad W(N, n_\uparrow) &= \frac{N!}{(n_\uparrow)!, (N - n_\uparrow)!} \ . \end{aligned}$$

(ii) What is the potential energy, $V(L)$, of the weight as a function of L?

$$V(L) = -M\mathrm{g}L = -M\mathrm{g}(n_\downarrow - n_\uparrow) \ .$$

Or we may write, as the only energy is potential:

$$L = -\frac{E}{M\mathsf{g}} \ .$$

(iii) Write down the canonical partition function. Evaluate it noting the analogy with the spin system in example 2 in the lectures.

$$W(N,E) = \sum_{n_\uparrow=0}^{N} \delta\left(E - \left(-\left[n_\downarrow - n_\uparrow\right]\right) M\mathsf{g}\right) W(N,n_\uparrow) \tag{2.2}$$

$$\begin{aligned}
\text{Then:} \quad Z(N,T) &= \int \mathrm{d}E\ W(N,E)\mathrm{e}^{-\beta E} \\
\text{(Use Eq. (2.2).)} \quad &= \sum_{n_\uparrow=0}^{N} \int \mathrm{d}E\ \delta\left(E - \left[n_\uparrow - n_\downarrow\right] M\mathsf{g}\right) W(N,n_\uparrow)\ \mathrm{e}^{-\beta E} \\
&= \sum_{n_\uparrow=0}^{N} W(N,n_\uparrow)\ \mathrm{e}^{-\beta M\mathsf{g}\left(n_\uparrow - n_\downarrow\right)} \\
&= \sum_{n_\uparrow=0}^{N} \frac{N!}{(n_\uparrow)!,(N-n_\uparrow)!} \mathrm{e}^{-\beta M\mathsf{g}\left(n_\uparrow - (N-n_\uparrow)\right)} \\
\text{(Recognise binomial.)} \quad &= \left(\mathrm{e}^{-\beta M\mathsf{g}} + \mathrm{e}^{+\beta M\mathsf{g}}\right)^N \\
&= \left(2\cosh[\beta M\mathsf{g}]\right)^N \qquad (2.3) \\
\Rightarrow \quad -k_{\mathrm{B}}T\ln Z = F(T,M\mathsf{g},N) &= -k_{\mathrm{B}}TN\ln(2\cosh[\beta M\mathsf{g}]) \ . \qquad (2.4)
\end{aligned}$$

(iv) Determine $\langle L\rangle$ as a function of T. By definition of an average, with a probability distribution $p(n_\uparrow)$:

$$\begin{aligned}
\langle L\rangle &= \sum_{n_\uparrow=0}^{N} (n_\downarrow - n_\uparrow) p(n_\uparrow) \\
&= \sum_{n_\uparrow=0}^{N} \left(n_\downarrow - n_\uparrow\right) W(N,n_\uparrow) \frac{\mathrm{e}^{\beta M\mathsf{g}\left(n_\downarrow - n_\uparrow\right)}}{Z} \\
&= \frac{1}{\beta}\frac{\partial}{\partial f} \ln Z(T,f,N)|_{f=M\mathsf{g}} \\
&= -\frac{\partial}{\partial f} F(T,f,N)\Big|_{f=M\mathsf{g}} \\
&= -\frac{\partial}{\partial f}\left\{-k_{\mathrm{B}}T\ln\left[2\cosh\beta f\right]^N\right\}\Big|_{f=M\mathsf{g}} \\
&= k_{\mathrm{B}}TN\beta\ \frac{2\sinh\beta f}{2\cosh\beta f}\Big|_{f=M\mathsf{g}} \\
&= N\tanh\beta M\mathsf{g} \\
\text{(Small mass, } M\text{.)} \quad &\simeq N\frac{M\mathsf{g}}{k_{\mathrm{B}}T} \\
\Rightarrow \quad \frac{\langle L\rangle k_{\mathrm{B}}T}{N} &= M\mathsf{g} \ .
\end{aligned}$$

Thus if we compare the last line with Hooke's law, noting the implicit sign of the for Mg, we see the spring constant $K = k_{\text{B}}T/N$, proportional to temperature. This is the signature of an entropic force - the desire of the chain to maximise entropy becomes greater for increasing temperature (as it is TS in the free energy).

(v) Which parts of the free energy are in balance at equilibrium?
The potential energy of the weight and the entropy of the chain.

♡*Problem 2.7* **Counterpart to equipartition for different potential shapes**
(A memento from Temperature and matter sheet 5.)
The aim is to derive the counterpart to $\frac{1}{2}k_{\text{B}}T$ per degree of freedom for potential energy of the form $V_n(q) = V_0|q|^n$, with $n > 0$ and $V_0 > 0$.

The corresponding partition functions, Z_n:

$$Z_n = \int_{-\infty}^{\infty} \mathrm{d}q \ \exp\left(-\frac{V_0|q|^n}{k_{\text{B}}T}\right) \ .$$

And the average value for the potential energy is

$$\langle V_n \rangle = \frac{\displaystyle\int_{-\infty}^{\infty} \mathrm{d}q \ V_0|q|^n \ \exp\left(-\frac{V_0|q|^n}{k_{\text{B}}T}\right) \ \mathrm{d}q}{\displaystyle\int_{-\infty}^{\infty} \mathrm{d}q \ \exp\left(-\frac{V_0|q|^n}{k_{\text{B}}T}\right)} \ .$$

We will break the derivation of this quantity into a number of stages.

The Boltzmann factor for $V_n(x)$ is

$$\exp\left(-\frac{V_0|q|^n}{k_{\text{B}}T}\right) \ .$$

So the *normalised* probability for the particle being $q_0 \leq q \leq q_0 + \mathrm{d}q$, $p(q)\mathrm{d}q$, is

$$p(q) \ \mathrm{d}q = \frac{\displaystyle\exp\left(-\frac{V_0|q|^n}{k_{\text{B}}T}\right) \ \mathrm{d}q}{\displaystyle\int_{-\infty}^{\infty} \mathrm{d}q \ \exp\left(-\frac{V_0|q|^n}{k_{\text{B}}T}\right) \ \mathrm{d}q} \ .$$

It is convenient to define the quantity Z_n:

$$Z_n = \int_{-\infty}^{\infty} \mathrm{d}q \ \exp\left(-\frac{V_0|q|^n}{k_{\text{B}}T}\right) \ .$$

And the average value for the potential energy is

$$\langle V_n \rangle = \int_{-\infty}^{\infty} \mathrm{d}q \ V_0|q|^n p(q) = \frac{\displaystyle\int_{-\infty}^{\infty} \mathrm{d}q \ V_0|q|^n \ \exp\left(-\frac{V_0|q|^n}{k_{\text{B}}T}\right) \ \mathrm{d}q}{\displaystyle\int_{-\infty}^{\infty} \mathrm{d}q \ \exp\left(-\frac{V_0|q|^n}{k_{\text{B}}T}\right) \ \mathrm{d}q} \ .$$

(i)

$$\begin{aligned}
-\frac{\partial \ln Z_n}{\partial \beta} &= -\frac{1}{Z_n}\frac{\partial Z_n}{\partial \beta} \\
&= -\frac{1}{Z_n}\frac{\partial}{\partial \beta}\int_{-\infty}^{\infty} \mathrm{d}q \ \exp\left(-\beta V_0 |q|^n\right) \ \mathrm{d}q \\
&= -\frac{1}{Z_n}\int_{-\infty}^{\infty} \mathrm{d}q \ \left[-V_0|q|^n\right] \exp\left(-\beta V_0 |q|^n\right) \ \mathrm{d}q \\
&= \langle V_n \rangle
\end{aligned}$$

(ii) If

$$y = (\beta V_0)^{1/n}\, q \quad \Rightarrow \quad \mathrm{d}q = \mathrm{d}y\, (\beta V_0)^{-1/n} \ .$$

Then:

$$\begin{aligned}
Z_n &= \int_{-\infty}^{\infty} \mathrm{d}q \ \exp\left(-\frac{V_0|q|^n}{k_{\mathrm{B}}T}\right) \ \mathrm{d}q \\
&= (\beta V_0)^{-1/n}\int_{-\infty}^{\infty} \mathrm{d}y \mathrm{e}^{-|y|^n} = (\beta V_0)^{-1/n}\, I_0
\end{aligned} \tag{2.5}$$

Note I_0 is a pure number.

(iii) Now we use Eq. (2.5)

$$\begin{aligned}
\ln Z_n &= -\frac{1}{n}\ln(\beta V_0) + \ln I_0 \\
\Rightarrow \quad -\frac{\partial \ln Z_n}{\partial \beta} &= \frac{1}{n}\frac{1}{\beta V_0} V_0 \qquad\qquad = \frac{1}{n} k_{\mathrm{B}} T \ .
\end{aligned}$$

For $n = 2$ we obtain the harmonic oscillator and get the usual Equipartition theorem.

(iv) For the "flat Earth" gravitational potential $V(z) = mgz$, we use $z \geq 0$ and get the same answer ie

$$\langle V_1 \rangle = k_{\mathrm{B}} T \ .$$

♣*Problem 2.8* **Entropic spring.**

Consider the system whose cross section is shown in Fig. (2.2). This system displays temperature-dependent competition between the internal energy of the (mechanical) spring and the entropy of the hinged rods.

The spring constant of the massless spring is K and it is unstretched when the rods are horizontal. The coupled rods (of length a and mass density ρ per unit length) are freely hinged at each end and at their join. Calculate the partition function of the system, and hence the average spring extension as a function of T.

Where does the entropy reside in this system?

Firstly calculate the moment of inertia of the rods as a function of the angle θ. The "vertical" displacement of the hinge between the rods is $\ell \sin\theta$ and the horizontal coordinate of the hinge is $\ell\cos\theta$, implying the spring is stretched to a length $2\ell(1-\cos\theta)$. So, defining I_0:

$$I(\theta) = 2\rho \int_0^{\ell\cos\theta} \frac{\mathrm{d}x}{\cos\theta}\,(x\sin\theta)^2 = \tfrac{2}{3}\rho\ell^3\cos^2\theta\sin^2\theta = I_0\cos^2\theta\sin^2\theta$$

The partition function is then, including rotational kinetic energy of the rods via their angular momentum, L, and making Z dimensionless by including $\hbar$:

$$Z = \int_0^{2\pi} \mathrm{d}\phi \int_{-\pi/2}^{\pi/2} \mathrm{d}\theta \, \sin\theta \exp\left(-\frac{K\ell^2 4(1-\cos\theta)^2}{2k_{\mathrm{B}}T}\right) \int_{-\infty}^{\infty} \frac{\mathrm{d}L}{\hbar} \, \exp\left(-\frac{L^2}{2Ik_{\mathrm{B}}T}\right)$$

The L integral yields $\sqrt{\pi 2IkT}/\hbar$, the ϕ integral yields 2π. Change variables to $u = \cos\theta$, then:

$$Z = 4\pi\sqrt{\frac{\pi k_{\mathrm{B}}T2I_0}{\hbar^2}} \int_0^1 \mathrm{d}u \, \sqrt{u^2(1-u^2)} \exp\left(-\frac{2K\ell^2}{k_{\mathrm{B}}T}(1-u)^2\right)$$

This result can probably be expressed, not very helpfully, in terms of incomplete Gamma and Beta functions. We will instead ask about the asymptotic behaviour at "low" and "high" temperatures.

Firstly, $T \to \infty$, i.e. $k_{\mathrm{B}}T \gg K\ell^2$. Then the part of the partition function associated with the spring is irrelevant as the exponent is zero (note the maximum elongation of the spring is 2ℓ which involves only a *finite* energy. Then (using the substitution $u^2 = t$ presently)

$$\begin{aligned} Z \simeq 4\pi\sqrt{\frac{\pi k_{\mathrm{B}}T2I_0}{\hbar^2}} \int_0^1 \mathrm{d}u \, \sqrt{u^2(1-u^2)} &= 2\pi\sqrt{\frac{\pi k_{\mathrm{B}}T2I_0}{\hbar^2}} \int_0^1 \mathrm{d}t \, (1-t)^{1/2} \\ &= 2\pi\sqrt{\frac{\pi k_{\mathrm{B}}T2I_0}{\hbar^2}} \frac{2}{3} \\ &= \frac{4\pi}{3}\sqrt{\frac{\pi k_{\mathrm{B}}T2I_0}{\hbar^2}} \ . \end{aligned}$$

We may write down the high-temperature free energy:

$$F = -k_{\mathrm{B}}T \ln Z \simeq -k_{\mathrm{B}}T \ln\left\{\frac{4\pi}{3} 2\pi\sqrt{\frac{\pi k_{\mathrm{B}}T2I_0}{\hbar^2}}\right\} \ ,$$

Now turn to $T \to 0$, noting that we require

$$\frac{\hbar^2}{2I_0} \ll k_{\mathrm{B}}T \ll K\ell^2 \ ,$$

i.e. $k_{\mathrm{B}}T$ is greater than the rotational level spacing, so not in quantum regime, but the spring is not significantly stretched by temperature. Now one cannot disregard the exponential, indeed it controls the part of the region of integration which makes the major contribution. This is near $u = 1$, so let $1 - u = x$ implying $u = 1 - x$:

$$\begin{aligned} Z &= 4\pi\sqrt{\frac{\pi k_{\mathrm{B}}T2I_0}{\hbar^2}} \int_0^1 \mathrm{d}u \, \sqrt{u^2(1-u^2)} \exp\left(-\frac{2K\ell^2}{k_{\mathrm{B}}T}(1-u)^2\right) \\ Z &= -4\pi\sqrt{\frac{\pi k_{\mathrm{B}}T2I_0}{\hbar^2}} \int_1^0 \mathrm{d}x \, (1-x)\sqrt{x-x^2} \exp\left(-\frac{2K\ell^2}{k_{\mathrm{B}}T}x^2\right) \\ &\simeq -4\pi\sqrt{\frac{\pi k_{\mathrm{B}}T2I_0}{\hbar^2}} \int_1^0 \mathrm{d}x \, \sqrt{2x} \exp\left(-\frac{2K\ell^2}{k_{\mathrm{B}}T}x^2\right) \ . \end{aligned}$$

Now let $t = x^2\ell^2 2K/(k_{\text{B}}T)$, then:

$$Z \simeq -2\pi\sqrt{\frac{\pi k_{\text{B}}T2I_0}{\hbar^2}}\left(\frac{k_{\text{B}}T}{2K\ell^2}\right)^{3/2}\int_{2K\ell^2/(k_{\text{B}}T)}^{0}\mathrm{d}t\, t^{1/4}\mathrm{e}^{-t}$$

(Note lower limit $\to \infty$.)
$$Z \simeq -2\pi\sqrt{\frac{\pi k_{\text{B}}T2I_0}{\hbar^2}}\left(\frac{k_{\text{B}}T}{2K\ell^2}\right)^{3/2}\int_{\infty}^{0}\mathrm{d}t\; t^{1/4}\mathrm{e}^{-t}$$

(Reverse limits, absorb -.)
$$= 2\pi\sqrt{\frac{\pi k_{\text{B}}T2I_0}{\hbar^2}}\left(\frac{k_{\text{B}}T}{2K\ell^2}\right)^{3/2}\int_{0}^{\infty}\mathrm{d}t\; t^{1/4}\mathrm{e}^{-t}$$

$$= 2\pi\sqrt{\frac{2K\ell^2\pi}{\hbar^2/2I_0}}\left(\frac{k_{\text{B}}T}{2K\ell^2}\right)^{2}\Gamma(5/4)\ .$$

The first factor has been rewritten to emphasise it is dimensionless.

And so the free energy is

$$F = -k_{\text{B}}T\ln Z \simeq -k_{\text{B}}T\left(2\ln\left\{\frac{k_{\text{B}}T}{2K\ell^2}\right\} + \text{constants}\right)$$

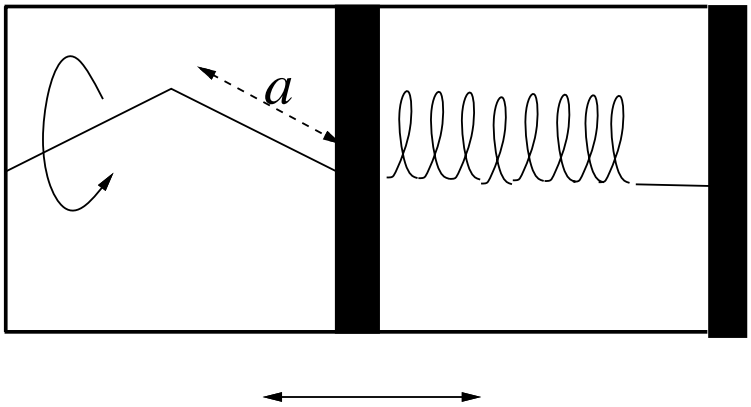

Figure 2.2: The entropic spring.